

Sunghwan Cho

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EDUCATION

University of Wisconsin-Madison

Sep 2022 – May 2027

B.S. Computer Engineering & Computer Science

Madison, WI

GPA: 3.93/4.00 | Dean's Honor List (All semesters)

Undergraduate TA, ECE 352: Digital System Fundamentals (SystemVerilog) | (2025 – Present)

EXPERIENCE

Computer Vision Research Intern

Sept 2025 – Present

Connected & Autonomous Transportation Systems Lab (Prof. Xiaopeng Li)

Madison, WI

- Implemented multi-camera 3D object detection pipelines in **Python** using **PyTorch** and **MMDetection3D**, preparing and partitioning standard datasets (DAIR-V2X, KITTI) for autonomous driving perception research
- Engineered end-to-end training and evaluation pipelines across multiple DAIR-V2X domains, performing annotation conversion to KITTI format and cross-domain benchmarking of **BEV/3D** detection using AP and IoU metrics
- Developed a single-view camera calibration pipeline using **PyTorch** and **OpenCV**, benchmarking deep learning based intrinsic estimation framework and achieving under **5%** focal length error on roadside infrastructure imagery

Software Engineer Intern

Dec 2025 – Jan 2026

PNS Networks

Seoul, South Korea

- Built a **full-stack** freight analytics platform using Python, Flask, React, and TypeScript, automating multi-year intermodal freight ETL pipelines, reducing manual processing by **95%**, enabling data-backed carrier renegotiation
- Analyzed **100k+** intermodal shipment records using **regression modeling** to compare operational variables and identify loss drivers, developing a continuous **risk-scoring model** to evaluate contract risk and high-loss scenarios

Technology Strategy & Transformation Intern

June 2025 – Aug 2025

Deloitte

Seoul, South Korea

- Initiated the digital transformation of a **\$50B** financial institution using **TOGAF enterprise architecture framework**, developing a modernization roadmap to improve operational & workflow efficiency across systems
- Assessed AI-OCR, blockchain, and eForm solutions through cross-industry benchmarking and requirements analysis, and defined a reference architecture by analyzing **Java**-based applications, designing inter-application integrations

PROJECTS

AI-Agent Productivity Platform | *Next.js, TypeScript, Spring Boot, FastAPI* | [GitHub](#)

Jan 2026 – Present

- Developed a full-stack AI platform with a chat interface where multiple LLM agents collaboratively debate and verify responses to reduce hallucination, using **Next.js**, **Spring Boot**, and **FastAPI**
- Implemented a multimodal ingestion pipeline including PDF parsing, OCR, layout-aware chunking, and vision grounding (**GPT-4o** / **Gemini Vision**) to enrich agent context beyond plain text
- Designed a microservice backend architecture with **Spring Boot** for product services and **FastAPI** for AI orchestration, implementing search-and-verify pipelines and structured run tracking using **PostgreSQL** (JSONB)

Bacterial Imaging UI for ELR-Based AST | *Python, Tkinter, OpenCV, Raspberry Pi*

Nov 2025 – Present

- Developed a Python-based imaging interface enabling real-time capture, preview, and automated storage of sessile droplet images to support next-generation **phenotypic antimicrobial susceptibility testing (AST)**
- Implemented a multi-tab UI with rendering, image scaling, and slide-level image management using **Tkinter**, **OpenCV**, and **Picamera2**, improving high-throughput screening workflows for up to 55 droplets per slide

Segway Self-Balancing Controller | *SystemVerilog, SAED 32nm* | [GitHub](#)

Aug 2025 – Jan 2026

- Built an FPGA-based real-time **control SoC** featuring PID control with **multi-sensor system integration** for gyro/accelerometer fusion, multi-protocol sensor interfaces (SPI, UART, A2D), and PWM outputs
- Achieved **timing closure at 333 MHz** by **pipelining** control datapaths and applying **synthesis constraints** using Synopsys Design Compiler on the **SAED 32nm** library, including post-synthesis timing analysis

TECHNICAL SKILLS

Languages: SystemVerilog, C, C++, Assembly, Java, Python, TypeScript, SQL, Go, Swift, MATLAB

Frameworks & Libraries: Next.js, React, Spring Boot, FastAPI, PyTorch, OpenCV, TanStack Query

Systems: PostgreSQL, WebSockets, REST APIs, MV3, RAG, Linux, Docker, Git, Synopsys Design Compiler